STAT 312: HW Assignment 06

Instructor: Dr. Md Mutasim Billah

Instructions: Complete all six problems. Show your work, include the requested sketches, and clearly identify your final answers.

Problem 1

The hourly wages in a particular industry are normally distributed with mean \$13.20 and standard deviation \$2.50. A company in this industry employs 40 workers, paying them an average of \$12.20 per hour. Is there sufficient evidence that this company pays substandard wages? Use a significance level of $\alpha = 0.01$.

Report H_0, H_a, the rejection region, the test statistic, and your conclusion in context.

Problem 2

For Jones's political poll, $n = 15$ voters were sampled. We wish to test

$$H_0 : p = 0.5 \qquad \text{versus} \qquad H_a : p < 0.5,$$

where p is the proportion of voters who favor Jones. The test statistic is Y, the number of sampled voters favoring Jones.

Hint: What is the distribution of Y? Assume independent responses, a fixed number of voters, two possible outcomes for each voter, and the same probability p of favoring Jones for each voter.

(a) Calculate α if the rejection region is RR $= \{y \leq 2\}$.

(b) Is our test equally good in protecting us from concluding that Jones is a winner if in fact he will lose? Suppose Jones will receive 30% of the votes ($p = 0.3$). Write an expression for the probability β that the sample will erroneously lead us to conclude that H_0 is true, and that Jones is going to win? You do not need to evaluate the expression.

Problem 3

Find the rejection region for each of the following hypothesis tests using the standard normal (Z) curve. For each part, identify the critical value(s) and draw a picture showing the rejection region.

(a)

$$H_0 : \mu = 72, \qquad H_a : \mu < 72, \qquad \alpha = 0.0721.$$

(b)

$$H_0 : \mu = 72, \qquad H_a : \mu > 72, \qquad \alpha = 0.005.$$

(c)

$$H_0 : \mu = 72, \qquad H_a : \mu \neq 72, \qquad \alpha = 0.126.$$

Problem 4

A vice president in charge of sales for a large corporation claims that salespeople average no more than 15 sales contacts per week. He would like to increase this figure. To check his claim, $n = 36$ salespeople are selected at random, and the number of contacts made by each is recorded for a single randomly selected week. The sample mean and sample variance of the 36 measurements are 17 and 9, respectively. Does the evidence contradict the vice president's claim?

Report H_0, H_a, the rejection region using $\alpha = 0.05$, the test statistic, and your conclusion in context.

Hint: Use the large-sample Z approximation with s estimating σ; do not use a t test for this problem.

Problem 5

The muzzle velocities of eight shells tested with a new gunpowder have sample mean $\overline{y} = 2959$ feet per second and sample standard deviation $s = 39.1$ feet per second. The manufacturer claims that the new gunpowder produces a mean velocity of at least 3000 feet per second. Do the sample data provide sufficient evidence to contradict the manufacturer's claim at the 0.025 significance level? Assume that the population of velocities is normally distributed.

(a) Report H_0, H_a, the rejection region using $\alpha = 0.025$, the test statistic, and your conclusion in context.

(b) Use the t table to place bounds on the p-value. Draw a picture showing the relevant area.

Problem 6

Use the t table to place bounds on the p-value for each of the following one-sample hypothesis tests. For each part, draw a picture showing the relevant area under the t curve.

(a) Right-tailed test: $n = 10$ and $T = 2.5$.

(b) Left-tailed test: $n = 7$ and $T = -1.20$.

(c) Two-tailed test: $n = 12$ and $T = 1.4$.

Problem 7

A machine in a factory must be repaired if it produces more than 10% defective items among the large lot of items it produces in a day. A random sample of 100 items from the day's production contains 15 defectives, and the supervisor says that the machine must be repaired. Does the sample evidence support this decision? Use a test with significance level $\alpha = 0.01$.

Report H_0, H_a, the rejection region, the test statistic, and your conclusion.